CICLOSPORIN

Prescribe by brand

Nephrotoxic — levels

CHEMICAL NAME Cyclic undecapeptide: cyclo[(E)-(2S,3R,4R)-3-hydroxy-4-methyl-2-(methylamino)oct-6-enoyl-L-2-aminobutyryl-N-methylglycyl-N-methyl-L-leucyl-L-valyl-N-methyl-L-leucyl-L-alanyl-D-alanyl-N-methyl-L-leucyl-N-methyl-L-leucyl-N-methyl-L-valyl] (cyclosporine)

CLASS Calcineurin inhibitor immunosuppressant

TRADE NAMES Neoral · Sandimmun · Capimune · Restasis (eye drops)

THERAPEUTIC USES / INDICATIONS

- Prevention of transplant rejection (kidney, liver, heart, bone marrow).
- Severe psoriasis, atopic dermatitis.
- Rheumatoid arthritis; nephrotic syndrome; acute severe ulcerative colitis.
- Dry eye disease (drops).

ADMINISTRATION

- **PO** twice daily (12 h apart) at the same times relative to meals; switching brands requires monitoring.
- **IV** infusion over 2–6 h (anaphylaxis risk with Cremophor) — use glass/non-PVC sets.
- Avoid grapefruit.

DOSAGES

- **Transplant:** 2–15 mg/kg/day in 2 doses, adjusted to trough levels (target range per transplant protocol).
- **Psoriasis/atopic dermatitis:** 2.5–5 mg/kg/day in 2 doses.
- **RA:** 2.5–4 mg/kg/day.

PHARMACOKINETICS

- **Peak:** 1.5–2 h. Variable bioavailability (microemulsion more consistent).
- Metabolised by CYP3A4; P-gp substrate.
- Excreted in bile. **t½:** 6–27 h.

NURSING MANAGEMENT

- Trough levels (before morning dose) as scheduled.
- Monitor BP, creatinine, K^+, Mg^{2+}, LFTs, lipids.
- Same brand each time.
- Oral hygiene for gum overgrowth; infection surveillance.

PHARMACODYNAMICS

- Binds **cyclophilin** → inhibits **calcineurin** → ↓ IL-2 transcription → blocks T-cell activation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Gingival hyperplasia, hirsutism, tremor.
- Hypertension, headache.
- GI upset; paraesthesia.

ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (acute and chronic).
- Hypertension; hyperkalaemia, hypomagnesaemia; hyperuricaemia.
- Infections; malignancy (skin, lymphoma).
- Neurotoxicity (PRES, seizures); hepatotoxicity; dyslipidaemia.

CONTRAINDICATIONS & CAUTIONS

- Uncontrolled hypertension; renal impairment (non-transplant indications).
- Malignancy (except non-melanoma skin cancer).
- Use with live vaccines; uncontrolled infections.

KEY INTERACTIONS

- **↑ levels (toxicity):** clarithromycin, erythromycin, azoles, diltiazem, verapamil, grapefruit.
- **↓ levels (rejection):** rifampicin, carbamazepine, phenytoin, St John's wort.
- **Nephrotoxic drugs (NSAIDs, aminoglycosides):** additive.
- **K^+-sparing diuretics:** hyperkalaemia; **statins:** myopathy.

PATIENT EDUCATION

- Take at the same times every day; do not switch brands.
- Avoid grapefruit and St John's wort; check all new medicines.
- Use sunscreen; report fever or infection.
- Keep dental hygiene.

KEY POINTS

- Calcineurin inhibitor — transplant immunosuppressant.
- Nephrotoxicity, hypertension, gum hyperplasia.
- Trough level monitoring; brand-specific.
- Major CYP3A4 interactions.

SOURCES CHECKED **[852]** Capimune 100 mg, soft capsules (UK SmPC) **[853]** SANDIMMUN Concentrate for Solution for Infusion 50mg/ml (UK SmPC) **[854]** Cyclosporine, oral (US label) **[855]** Sandimmune, IV, oral (US label)

Full references p. 432–457